

tools and libraries

Documentation: Tools and Libraries

This project leverages a combination of professional astronomical software and a suite of powerful Python libraries to form a complete scientific analysis pipeline.

External Software

HEASOFT & XSPEC

- **What It Is:** HEASOFT (High Energy Astrophysics Science Archive Research Center Software) is a comprehensive package of tools for high-energy astrophysics. Its most famous component is `XSPEC`, which is the global standard for X-ray spectral fitting.
 - **Primary Role in this Project:** `XSPEC` was used for the most critical scientific modeling step: **calibrating the solar spectrum**.
 - **Why It Was Essential:** `XSPEC` is designed to perform **forward-fitting**. It understands how to use instrumental calibration files (RMF and ARF) to simulate the "blur" and "filter" of a detector. This allowed us to take the raw XSM data and determine the true, physical X-ray spectrum of the Sun, `F_sun(E)`, by removing the instrument's unique distortions.
 - **How It Was Used:** We used `XSPEC`'s Python interface, `pyxspec`, inside the `run_analysis_batch.py` script to automate this calibration for every observation.
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Core Python Libraries

`pathlib`

- **What It Is:** A modern, built-in Python library for handling file system paths in an object-oriented and operating-system-independent way.
- **Primary Role in this Project:** Used in every script to define, manage, and manipulate all file and directory paths (e.g., `RESULTS_DIR =`

`Path("/home/saurabh/moon_project/results/batch_run/")`). It makes the code cleaner and less prone to errors related to file paths.

pandas

- **What It Is:** The most popular data manipulation and analysis library in Python, centered around a powerful data structure called the DataFrame.
- **Primary Role in this Project:** `pandas` was the backbone of our data handling in Phase 1.
- **Specific Uses:**
 - Reading and parsing the large CSV files created by the metadata extraction scripts (`perMonthEachDay_extract_class_metadata.py` , `new_extract_xsm_metadata.py`).
 - Performing the crucial time-based filtering and merging in `match_CLASS_XSM.py` .
 - Reading the final list of observations to be processed in `run_analysis_batch.py` .
 - Creating and saving the final `final_abundance_ratios.csv` file.

numpy

- **What It Is:** The fundamental library for numerical computing in Python. It provides the powerful `ndarray` (N-dimensional array) object and a huge collection of mathematical functions to operate on these arrays.
- **Primary Role in this Project:** `NumPy` handled almost all numerical calculations.
- **Specific Uses:**
 - Storing spectral data (energy, counts) as arrays.
 - Performing array mathematics, such as summing the 1-second XSM spectra in `prepare_xsm_for_xspec.py` .
 - The **rebinning** operation (`.reshape(...).sum(...)`) was a key NumPy feature.
 - Calculating the final abundance ratios from the measured line fluxes.

astropy

- **What It Is:** The core community-developed library for professional astronomy in Python.
- **Primary Role in this Project:** Handling astronomy-specific file formats and data types.
- **Specific Uses:**
 - **FITS File I/O:** The `astropy.io.fits` module was used in every script that needed to read from or write to a `.fits`, `.pha`, `.arf`, or `.rmf` file.
 - **Time Objects:** The `astropy.time.Time` object was used in `prepare_xsm_for_xspec.py` to perform robust and accurate comparisons between the observation timestamps.

scipy

- **What It Is:** A library for scientific and technical computing, built on top of NumPy.
- **Primary Role in this Project:** Used for its powerful scientific fitting and interpolation capabilities.
- **Specific Uses:**
 - **Interpolation:** `scipy.interpolate.interp1d` was used in `py_calibrate_xsm.py` to turn the discrete points of the unfolded solar spectrum into a continuous function, `F_sun(E)`.
 - **Optimization/Fitting:**
 - `scipy.optimize.minimize` was used in `py_forward_model_and_subtract.py` to find the best-fit `alpha` (scatter scaling factor) for the continuum model.
 - `scipy.optimize.curve_fit` was used in `py_fit_xrf_lines.py` to perform the final multi-Gaussian fit to the residual XRF peaks.

matplotlib

- **What It Is:** The most widely used library for creating static, publication-quality plots and visualizations in Python.
- **Primary Role in this Project:** Used for creating all the 2D diagnostic plots.
- **Specific Uses:**

- Visualizing the unfolded solar spectrum in `py_calibrate_xsm.py` .
- Creating the crucial diagnostic plot (continuum fit + residual) in `py_forward_model_and_subtract.py` .
- Plotting the final Gaussian line fits in `py_fit_xrf_lines.py` .
- Used by `healpy` as the backend for generating and saving the final 2D maps.

healpy

- **What It Is:** The primary Python library for working with **HEALPix** (Hierarchical Equal Area isoLatitude Pixelation) data.
- **Primary Role in this Project:** Used to create the final, scientifically accurate 2D maps.
- **Specific Uses:**
 - The `hp.ang2pix` function was used to convert latitude/longitude coordinates into HEALPix pixel indices.
 - The `hp.mollview` function was used in `plot_healpix_map.py` to generate the final **Mollweide projection** map, which is an equal-area representation of a sphere.

plotly

- **What It Is:** A modern library for creating interactive, web-based data visualizations.
- **Primary Role in this Project:** Used to create the interactive 3D map of the Moon.
- **Specific Uses:**
 - The `go.Scatter3d` object was used in `plot_3d_map.py` to plot each observation as a point on a 3D sphere.
 - The `hovertemplate` feature was used to add interactive tooltips showing the abundance data for each point.
 - The final output was an HTML file that can be opened in any web browser.